Requirements Gathering

1. Stakeholder Analysis

The enterprise analytics platform serves four executive and operational leadership roles. Rather than viewing data through isolated transactional metrics, each stakeholder approaches the system with distinct strategic goals, operational pressures, and resource-allocation responsibilities.

- **Chief Executive Officer (CEO):** Overseeing multi-national commercial operations across seven countries, the CEO needs macro-level clarity on top-line revenue velocity, market-level profitability, and customer equity growth. Their primary challenge is evaluating regional momentum across North America, Europe, Asia-Pacific, and the Middle East without getting overwhelmed by transactional noise. They need to identify which geographic territories demonstrate compounding returns on invested capital so they can direct expansion funds toward high-margin, high-lifetime-value regions while restructuring or consolidating underperforming operations.
- **Chief Operating Officer (COO):** Leading reverse logistics, supply chain integrity, and warehouse fulfillment across nineteen regional hubs, the COO contends with margin erosion caused by a 6.1% return rate, a 6.6% pre-shipment order cancellation rate, and backlog accumulation. Their core need is operational accountability: isolating controllable warehouse errors such as wrong item dispatches, packaging transit damage, and dispatch delays from natural buyer remorse. They require immediate visibility into vendor defect rates across ten suppliers, warehouse pick bottlenecks, checkout payment gateway drop-offs, and shipping subsidization costs to protect customer trust and delivery service-level agreements (SLAs).
- **Chief Financial Officer (CFO):** Safeguarding enterprise unit economics and bottom-line profit margins, the CFO faces severe margin leakage caused by unmanaged promotional discounting and coupon code stacking. Their critical concern is identifying where gross revenue evaporates between list price and realized net income. They need empirical guardrails on promotional spend, line-item visibility into cost of goods sold (COGS) across product categories, and clear customer acquisition cost (CAC) payback periods across ten marketing channels to ensure growth capital is allocated only to channels delivering durable customer equity.
- **Head of Customer Retention (CRM Lead):** Managing post-purchase customer journeys and lifecycle engagement, the Retention Lead confronts steep attrition cliffs where first-

time buyers fail to transition into repeat brand advocates. They must protect high-equity VIP and Premium customer accounts from defection, understand how fulfillment delays degrade re-order rates, and execute win-back campaigns that do not sacrifice gross margins. They need a system that identifies churn risk early, leverages dormant loyalty point equity as a zero-discount incentive, and filters out unprofitable serial returners before retention budgets are spent.

2. User Stories & Use Cases

CEO Scenario: International Capital Allocation and Regional Expansion

- **User Story:** As the Chief Executive Officer, I want an executive spatial overview linking revenue scale, realized profit margins, and long-term customer equity across operating territories, so that I can deploy international capital toward accelerating markets while consolidating low-margin operations.
- **End-to-End Operational Scenario:** Entering the executive command center on Monday morning ahead of a board strategy meeting, the CEO opens the international performance view to resolve whether capital should be channeled into opening new local distribution centers in the United Kingdom and Germany or expanding existing infrastructure in the United States.

The CEO begins by scanning the interactive world map, observing that while the United States generates the vast majority of gross volume (59.2% of orders and $754.8K in Net Sales), international markets are sustaining healthy margins between 44% and 46%. To assess long-term momentum, the CEO checks the five-year regional sales trajectory, verifying that both the European and North American East operating regions have maintained compounding multi-year acceleration.

To determine true capital efficiency, the CEO inspects the country-level unit economics quadrant comparing Customer Acquisition Cost against Customer Lifetime Value. Here, the strategic choice becomes clear: the United Kingdom ($10,083 average CLV) and Germany ($9,724 average CLV) outperform against an acquisition cost of approximately $41, placing them squarely in the high-equity, low-cost quadrant. In contrast, the United Arab Emirates averages a lower lifetime value ($8,154 CLV) with an equivalent acquisition cost.

Drilling down further, the CEO evaluates the customer tier distribution across these territories, noting that Consumer accounts represent over 55% of transactions, while VIP and Business buyers generate disproportionately higher margins. Finally, the CEO examines the top ten profit-generating cities to identify urban commercial centers suitable for local

inventory staging. Armed with this unified picture, the CEO formulates an investment directive: ring-fence 20% incremental expansion funding for regional fulfillment hubs in top European urban centers while mandating that mature domestic markets maintain a 45% realized profit margin threshold.

COO Scenario: Root-Cause Return Auditing and Fulfillment Bottleneck Remediation

- **User Story:** As the Chief Operating Officer, I want an operational diagnostic flow tracking returns, supplier defect rates, warehouse queues, and shipping cost distributions, so that I can eliminate controllable fulfillment errors and unblock unshipped orders before service-level agreements are breached.
- **End-to-End Operational Scenario:** During daily fulfillment operations review, the COO investigates why reverse logistics expenses are running above plan while customer service reports elevated inquiries regarding pending orders. Navigating to the operational auditing interface, the COO executes a two-stage triage covering supplier defect control and fulfillment flow recovery.

In the first stage, the COO investigates product returns to separate internal operational failures from customer preference changes. Inspecting the ranked return reasons, the COO discovers that nearly half of all returns (49.2%) stem from controllable operational defects: incorrect items dispatched (16.4%), delivery transit delays (14.8%), packaging transit damage (13.1%), and manufacturing product defects (4.9%). To hold external partners accountable, the COO inspects the supplier-by-category quality matrix, instantly spotting that specific vendor and product combinations exceed the 8% return threshold. Correlating this against warehouse performance data reveals that fulfillment centers WH-004 and WH-005 account for the highest concentration of delayed and damaged dispatches. The COO immediately issues formal Supplier Corrective Action Requests (SCAR) to underperforming vendors and mandates air-cushion dunnage packaging upgrades at the offending warehouses.

In the second stage, the COO transitions to unblocking the fulfillment pipeline. Reviewing the order flow telemetry, the COO notes that 16.7% of all received orders fail to complete cleanly due to cancellations, returns, and stuck queues. Examining the warehouse backlog visual reveals an urgent operational risk: a cluster of orders marked as fully paid are sitting in pending pick queues past their twelve-hour dispatch window. To prevent order cancellations caused by customer impatience, the COO dispatches an immediate priority wave pick alert to warehouse supervisors. Finally, reviewing shipping cost distributions across delivery tiers, the COO notices that small-basket Same Day and Express orders are eating up 15% to 18% of

line-item net sales in subsidized freight. The COO establishes an operational policy: enforce a strict $150 minimum order threshold on expedited shipping methods while raising warehouse safety stock reorder points by 25% on the top ten revenue-losing backordered items.

CFO Scenario: Promotional Margin Defense and Acquisition Capital Reallocation

- **User Story:** As the Chief Financial Officer, I want to trace the complete financial bridge from gross revenue to bottom-line profit while analyzing promotional price elasticity and channel payback ratios, so that I can eliminate value-destroying discount promotions and shift acquisition spend toward high-return channels.
- **End-to-End Operational Scenario:** During quarterly budget reconciliation, the CFO seeks to understand why healthy top-line gross sales growth of 14% is not translating into proportional bottom-line cash generation. The CFO uses the financial governance interface to audit promotional discounting and calibrate customer acquisition budgets.

The CFO begins by reviewing the per-order financial waterfall bridge, which visualizes the stepwise erosion of revenue: an average gross order value of $1,365 is reduced by $252 in discounts, offset by $162 in tax and shipping collections to reach $1,275 in net sales, from which $648 in product COGS and $35 in outbound logistics are deducted, leaving $591 in realized net profit. To identify where margin leakage is concentrated, the CFO inspects the discount rate margin curve. The data reveals an immediate financial danger zone: orders discounted up to 15% maintain healthy margins of 51%, but discounts exceeding 35% collapse margins down to 20.1%, with discounts above 40% producing a 20.6% negative-profit incidence.

Drilling into specific promotional mechanisms, the CFO inspects coupon tier performance and discovers that campaigns utilizing deep codes (SAVE35 through SAVE49) consistently drag profit margins below the enterprise 30% safety floor. Furthermore, a segmentation check shows that repeat VIP and Loyal buyers are receiving deep percentage discounts on purchases they would have made at full catalog price.

To reallocate growth capital, the CFO examines marketing channel payback, ranking acquisition streams by their CLV-to-CAC ratios. Channels such as Referral, Organic Search, and Email deliver high ratios above 220x, while certain paid social and display channels struggle to generate compounding repeat orders. Armed with these findings, the CFO establishes two executive financial guardrails: configure an automated 30% discount ceiling in the checkout engine, retire all coupon tiers above SAVE30, and reallocate 25% of

acquisition budgets away from low-return channels toward top-performing retention co-funding at the critical Order #3 to Order #4 customer transition point.

Customer Retention Lead Scenario: Churn Intervention and Profitable Win-Back

- **User Story:** As the Customer Retention Lead, I want to track customer lifecycle attrition cliffs, correlate fulfillment friction with churn risk, and identify dormant loyalty point balances, so that I can deploy zero-discount re-engagement campaigns to protect high-value customer equity without eroding product margins.
- **End-to-End Operational Scenario:** With quarterly customer attrition creeping toward targeted limits, the Retention Lead opens the churn command center to pinpoint when and why customers disengage and to execute a margin-safe win-back campaign.

The Retention Lead starts by inspecting the cohort retention triangle across multi-year purchasing cohorts. The progression reveals two severe drop-off cliffs: an initial 28% drop between Order #1 and Order #2, and a second 21% attrition drop between Order #3 and Order #4, after which surviving customers reliably compound into loyal buyers generating between $9,000 and $27,000 in lifetime value. To understand what triggers early customer drop-off, the Retention Lead analyzes the post-order churn spline mapped against fulfillment service delivery gaps. The correlation is striking: while on-time deliveries exhibit an attrition baseline of 11.8%, customer churn risk jumps sharply to 32.5% when delivery delays reach two days and spikes past 48% when delays are combined with product damage or returns.

To design a win-back campaign that does not erode margins, the Retention Lead evaluates the win-back offer sensitivity matrix. Standard industry practices rely on sending 35% discount coupons to lapsing buyers, but the matrix proves that this tactic destroys profitability, dropping realized net margins to 15.9%. Instead, the Retention Lead looks at the customer loyalty balance metrics, which show that 47.9% of earned loyalty points remain unredeemed across customer accounts. The sensitivity matrix shows that a zero-discount outreach reminding buyers of their unredeemed points preserves a strong 50.4% to 51.8% net margin across all customer segments.

Finally, the Retention Lead accesses the prioritized CRM action queue, which surfaces high-CLV At-Risk customer accounts sorted by available equity. The queue automatically excludes unprofitable serial returners whose cumulative account margin is negative. Verifying the top VIP and Premium accounts, their unredeemed point balances, and their preferred contact channels, the Retention Lead triggers the outreach queue, instantly dispatching targeted, zero-discount loyalty reminders to protect enterprise customer equity while preserving bottom-line profit.

3. Functional Requirements:

- **Role-Tailored View Navigation:** The system shall provide dedicated executive views for the CEO, COO, CFO, and Customer Retention Lead, ensuring each dashboard displays role-specific decision banners, strategic rules, and analytical visualizations without requiring multi-page application reloading.
- **Multi-View Decision Sub-Routing:** The system shall support secondary operational decision buttons within complex functional domains, allowing the COO to toggle between supplier return audits and warehouse fulfillment bottlenecks, and the CFO to switch between promotional discount guardrails and marketing acquisition economics.
- **Global Dimensional Slicing:** The system shall provide persistent global filtering controls across geographic countries, customer tiers, and transaction years, instantly recomputing all dependent metrics and visualizations upon user selection.
- **Filter State Synchronization and Reset:** The system shall maintain bidirectional filter synchronization across all UI components—including interactive spatial maps, dropdown selectors, and metric cards—and include a single-click reset function restoring all filters to the full enterprise dataset.
- **Dynamic Order Count and Coverage Indicator:** The system shall dynamically calculate and display an active transaction counter indicating the exact number and percentage of records reflected in the current filter state relative to the total dataset.
- **Interactive Spatial Geographic Mapping:** The system shall render an interactive choropleth world map displaying country revenue scale, margin health, and order counts, supporting click-to-filter geographic selection across all seven operating territories.
- **Financial Profit and Loss Bridge Visualization:** The system shall compute and visualize an order-level financial waterfall bridge tracking the step-by-step transition from gross sales through promotional discounts, tax and shipping additions, product cost of goods sold, and final realized net profit.
- **Promotional Discount and Margin Tracking:** The system shall plot average realized profit margins across discrete promotional discount bands alongside a static thirty-percent enterprise target floor, highlighting where promotional depth leads to margin collapse.

- **Supply Chain Tracking:** The system shall trace the proportional flow of orders from initial payment statuses (Paid, Pending, Failed, Refunded) into final order fulfillment outcomes (Completed, Cancelled, Returned, Pending) to identify operational drop-off points.
- **Controllable Operational Return Classification:** The system shall isolate and quantify controllable reverse logistics failure reasons (wrong product shipped, transit damage, delivery delay, product defect) from buyer preference returns, providing category and supplier cross-tabulations.
- **Warehouse Fulfillment:** The system shall categorize and visualize pending and cancelled order volumes across all nineteen regional fulfillment warehouses, isolating orders marked as paid that remain unshipped beyond operational thresholds.
- **Retention Lifecycle Analysis:** The system shall construct an acquisition cohort retention matrix tracking customer survival rates across order milestones from initial purchase to repeat champion status, highlighting specific drop-off cliffs.
- **Fulfillment Delay Churn Correlation:** The system shall analyze and display the mathematical relationship between delivery service-level agreement delay days and post-purchase customer churn probabilities, distinguishing standard orders from defect-impacted shipments.
- **Prioritized CRM Retention:** The system shall generate an exportable, ranked action table of high-value At-Risk customer accounts displaying customer identifiers, tiers, lifetime value equity, unredeemed loyalty point balances, and preferred marketing channels while automatically filtering out customers with negative cumulative net profit.
- **Campaign and Coupon Tier Financial Auditing:** The system shall cross-reference individual marketing campaigns and coupon code tiers against realized net margins, highlighting promotional campaigns where total discount giveaway exceeds fifty percent of net profit.
- **Shipping Cost Distribution Analysis:** The system shall compute shipping costs as a percentage of net sales across expedited, express, standard, and economy freight tiers, identifying basket-size thresholds where subsidized delivery erodes transaction margins.

4. Non-Functional Requirements:

- **Performance and Rendering Latency:** The system shall complete all dataset filtering, statistical recalculations, and visual chart rendering within 150 milliseconds following user input or slicer adjustment on standard desktop client hardware.
- **Viewport Fit and Spatial Ergonomics:** The complete analytical workspace—including navigation headers, headline metric ribbons, primary strategic visual zones, and diagnostic charts—shall display harmoniously within a standard 16:9 widescreen display (1920x1080) without requiring vertical document scrolling.
- **Pre-Attentive Visual Hierarchy:** Visual typography shall maintain strict hierarchical contrast where headline KPI metrics appear at least twice the scale of supporting benchmark labels, enabling executive stakeholders to perform rapid at-a-glance health assessments within five seconds of view selection.
- **Semantic Color Standardization:** All chart and metric encodings shall strictly enforce a cohesive four-color semantic system: deep navy and blue for transaction volume and gross scale, teal and emerald for profit margins and customer retention, amber for pending queues and at-risk warnings, and coral red for defects, cancellations, and financial leakage.
- **Accessibility and Contrast Compliance:** All text elements, chart data labels, and interactive table rows shall maintain a minimum color contrast ratio of 4.5:1 against their adjacent container backgrounds, adhering to WCAG AA accessibility standards across both light interface cards and dark cartographic surfaces.
- **Mathematical Integrity and Division Safeguards:** All ratio, margin, and percentage calculations across sparse or heavily filtered dataset slices shall incorporate protective denominator fallbacks, guaranteeing zero divide-by-zero exceptions, missing numbers, or invalid floating-point values.
- **Reconciliation Accuracy:** Across any valid combination of geographic, customer, or temporal filters, the mathematical sum of line-item sales and net profits displayed within visual charts shall reconcile exactly with the aggregate headline figures in the KPI ribbon within a tolerance of one dollar.